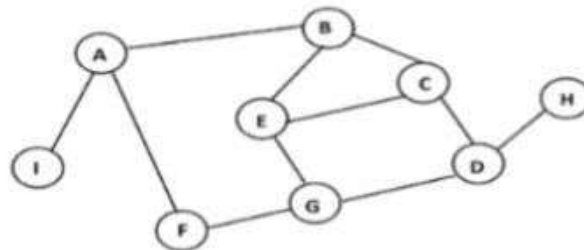


Course: CSE 207- Data Structures, Section-1
Instructor: Dr. Maheen Islam, Associate Professor CSE Department
Full Marks: 50
Time: 1 Hour and 15 Minutes

Note: There are FIVE questions, answer ALL of them. Course Outcome (CO), Cognitive Level and Mark of each question are mentioned at the right margin.

1. a. Show breadth-first traversal of the following graph, where H is the starting node. [CO2,C1, Mark: 7]
 Visit adjacent nodes in an anticlockwise ordering from a particular node (12 o'clock position).



[CO2,C1,
Mark: 3]

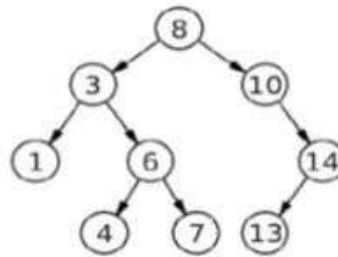
- b. Draw the adjacency list representation of the above graph.
2. Consider the following code in C. The function *Fun1* is called from the *main* function which receives the address of the root of a binary search tree. *Fun1* is called recursively. Consider the address of nodes from the table below. Now Show the output of the following BST using the below code by providing step by step output of the recursive function.

Node	Address	Node	Address	Node	Address
8	AA	1	EE	7	JJ
3	BB	6	FF	14	OO
10	CC	4	DD	13	TT

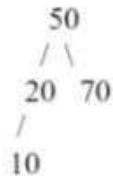
```
Fun1(tree)
{
    1.for d = 1 to height(tree)+1
        1.1. Fun2(tree, d);
}
Fun2(tree, level)
{
    1.if tree is NULL then return;
    2.if level is 1, then
        2.1 print(tree->data);
    3.else if level greater than 1, then
        3.1 Fun2(tree->left, level-1);
        3.2 Fun2(tree->right, level-1);
}
```

```
struct Node
{
    int data;
    Node * right;
    Node * left;
};
```

}

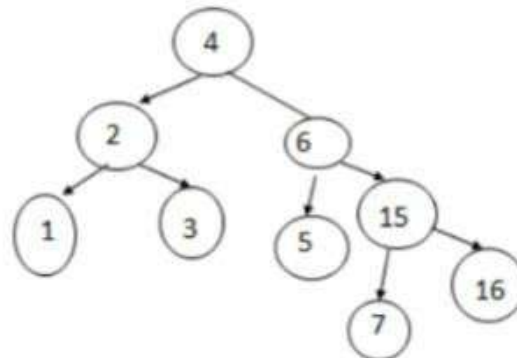


3. Write a program that will count and display the total number of children of each node in BST (Write only the Function). For example, for the following BST, output will be: [CO2,C3, Mark: 10]



Child of 50 is 2: 20 and 70
 Child of 20 is 1: 10
 Child of 10 is 0
 Child of 70 is 0

4. a. Construct an AVL tree using the data 15, 20, 24, 13, 7, 36, 25 and 50 entered as a sequential set. Clearly show the tree generated from each insertion. Also show the rotation that needs to be performed to balance the AVL tree. [CO3,C3, Mark: 7]
- b. Does the following BST is considered as an AVL tree? If it is, then delete the node containing 4 from the following tree. Draw each deleted scenario while you delete 4 from the below tree and make sure that, the tree created after deletion must be balanced. [CO3,C3, Mark: 3]



5. Construct the minimum spanning tree for the following graph. Also compute the weight. [CO2,C1, Mark: 10]

